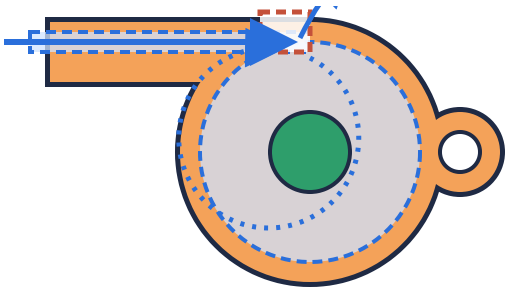
Project 2 | TinkerCad | About 45 min

Make a 3D Printed Whistle in TinkerCad

Design a whistle that really works, with a little “pea” rattling around inside it. It prints in one piece!



How does a whistle work?

You blow air into a **thin tunnel**. The air shoots out and hits a **sharp edge** at the sound window. The air splits and makes the air inside the round **chamber** spin and shake really fast. Fast shaking air = **sound!** The **pea** bounces around inside and makes the trilling “prrrrr” sound.

Here’s the plan

A whistle is a **round barrel** plus a **mouthpiece**, with the inside scooped out using **hole** shapes. You’ll build it from 7 simple parts, then group them into one whistle.

Hole
A see-through shape. When you group it with a solid shape, it cuts that space away.

Group
Joins shapes into one. This is when the holes actually get cut.

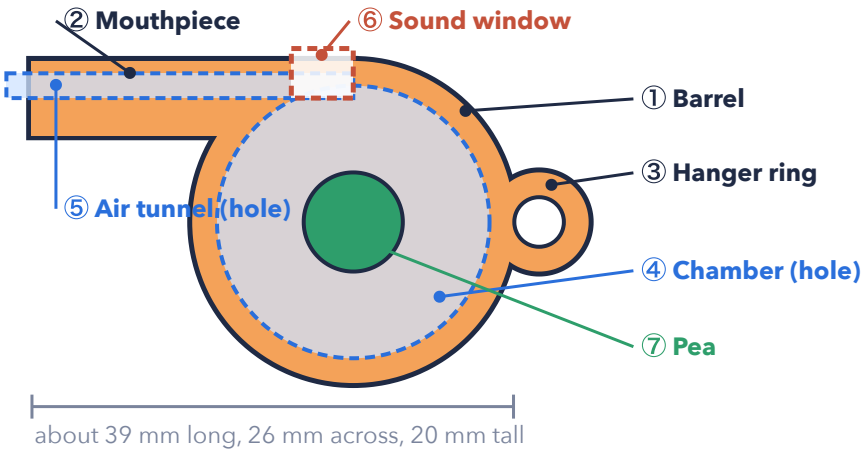
Align
Lines up the edges or centers of shapes so they fit together perfectly.

NAME YOUR DESIGN!
Your teacher downloads everyone’s whistles from TinkerCad to print them. Name yours **Whistle - First Name Last Initial** (like **Whistle - Maya R**) so we know it’s yours. You’ll do this in Step 1.

GOLDEN RULE
Made a mistake? Press `Ctrl` + `Z` to undo. You can’t break anything. 😄

🗺️ The Parts Map

This is your whistle seen from the **top**. Solid parts are orange. **Holes** are blue and dashed (the sound window is red so you can spot it).



	PART	SHAPE	SOLID OR HOLE	SIZE (MM)	HEIGHT OFF GROUND
①	**Barrel**	Cylinder	Solid	`26 × 26 × 20`	`0`
②	**Mouthpiece**	Box	Solid	`26 × 6 × 20`	`0`
③	**Hanger ring**	Cylinder	Solid	`8 × 8 × 4`	`0`
	...ring hole	Cylinder	**Hole**	`4 × 4 × 10`	`0`
④	**Chamber**	Cylinder	**Hole**	`22 × 22 × 16`	`2`
⑤	**Air tunnel**	Box	**Hole**	`28 × 2 × 16`	`2`
⑥	**Sound window**	Box	**Hole**	`5 × 4 × 12`	`4`
⑦	**Pea**	Sphere	Solid	`8 × 8 × 8`	`1.9`

HOW TO READ THE SIZES

Sizes go **long × wide × tall**. Looking from the top, "long" is left-right on your screen and "wide" is up-down. "Height off ground" is how high you lift the shape with the black cone. Keep this page next to you!

1 Get set up

- [ ] Open TinkerCad and click **Create → 3D Design**.
- [ ] **Name it right away!** Click the silly random name at the top left, type **Whistle – First Name Last Initial**, and press `Enter`.
- [ ] Check that **Snap Grid** (bottom right) says **`1.0 mm`**. Then click **Top** on the view cube so you're looking straight down.

Two tricks you'll use a lot

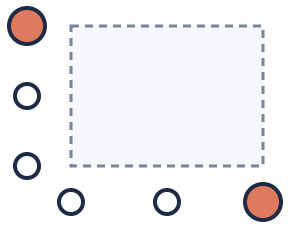
Typing a size: click a shape, hover over a white corner square, click a number that pops up, type the new number and press `Enter`. The **height** number is on the white square on top.
Lifting a shape: grab the **black cone** on top of the shape, click the number and type how high it goes.

2 The barrel ①

- [ ] Drag a **Cylinder** onto the middle of the workplane.
- [ ] Make it **`26 × 26 × 20`**.

3 The mouthpiece ②

- [ ] Drag a **Box** onto the workplane, to the **left** of the barrel and a little **lower** on your screen.
- [ ] Make it **`26 × 6 × 20`** (26 long, 6 wide, 20 tall).
- [ ] The box is still selected. Hold `Shift` and click the barrel, so both are selected.
- [ ] Click **Align** (or press `L`). Click the **right** dot in the left-right row, and the **top** dot in the up-down row (see the picture).
- [ ] Click the box by itself and press `←` **13 times**.



Click the two orange dots: **right** and **top**

CHECK
Look at the Parts Map. The back of the mouthpiece lines up with the back of the barrel, and its right end stops at the **middle** of the barrel.

4 The hanger ring ③

- [ ] Drag a **Cylinder** onto the empty workplane and make it `8 × 8 × 4`.
- [ ] Drag another **Cylinder** onto it, click **Hole** (◌), and make it `4 × 4 × 10`. Press `D` to drop it down to the workplane.
- [ ] Select both, **Align** both **middle** dots, then **Group** (`Ctrl` + `G`). Now it's a ring!
- [ ] Drag the ring onto the barrel and press `D` to drop it down. Shift-click the barrel, **Align** both **middle** dots.
- [ ] Click the ring by itself and press `→` **15 times**. It now sticks out the back of the barrel.

5 The chamber ④

- [ ] Drag a **Cylinder** onto the barrel. Click **Hole**. Make it `22 × 22 × 16`.
- [ ] Shift-click the barrel, **Align** both **middle** dots.
- [ ] Click the hole by itself and set its height off the ground to `2`. Now the barrel will have 2 mm thick walls all around.

CHECK

From the top, the ring sticks out the back of the barrel (the side away from the mouthpiece), and the chamber's dashed circle sits right in the middle of the barrel with an even border all around.

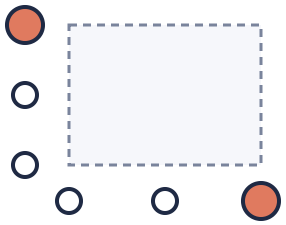
ALIGN TIP

Align should move the **new** shape, not the big one. If the wrong shape moves, press `Ctrl` + `Z`, move the new shape so it sits more on top of the big one, and try again.

6 The air tunnel ⑤ and sound window ⑥

These two little holes are what make the whistle whistle, so be careful here!

- ☐ **Air tunnel:** drag a **Box** onto the mouthpiece, click **Hole**, and make it `28 × 2 × 16`. Move it so it sticks out past the open (left) end, but **not** past the middle of the barrel.
- ☐ Shift-click the mouthpiece. **Align**: **right** dot and **top** dot.
- ☐ Click the tunnel by itself. Press ↓ **1 time**, then set its height off the ground to `2`.
- ☐ **Sound window:** drag a **Box** onto the mouthpiece close to the barrel, click **Hole**, and make it `5 × 4 × 12`.
- ☐ Shift-click the mouthpiece. **Align**: **right** dot and **top** dot.
- ☐ Click the window by itself. Press ↑ **1 time**, then set its height off the ground to `4`.

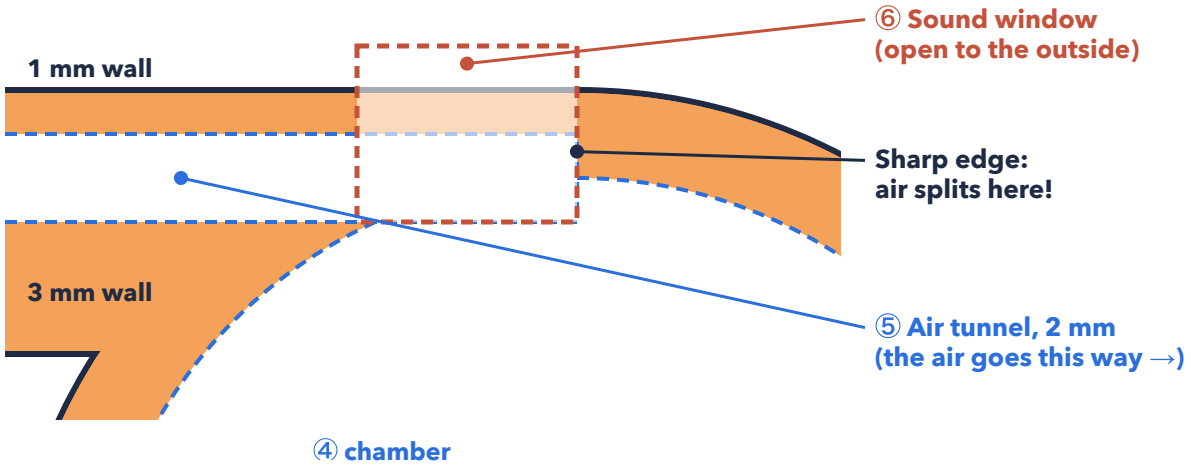


Same dots for both: **right** and **top**

WHY DOES THIS WORK?

The tunnel is only 2 mm tall, so the air comes out as a thin, fast stream. The window sits right where the tunnel meets the barrel, so the stream hits the sharp edge there and splits in half.

Close-up: how it should look from the top



7 Group to cut the holes

- [ ] Press `Ctrl` + `A` to select everything.
- [ ] Click **Group** (or `Ctrl` + `G`).

CHECK

Click the view cube's **Home** button (the little house) and drag the view cube to look around:

- Look into the end of the mouthpiece: you should see a **thin slot** (the air tunnel).
- On the flat top of the mouthpiece, right next to the barrel, you should see a **rectangle opening** (the sound window).

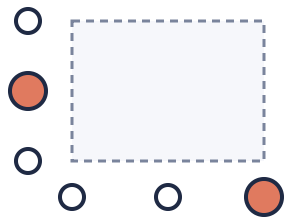
UH OH?

No slot or no window? Press `Ctrl` + `Z` to undo the group, then check the hole sizes, heights and arrow presses against the Parts Map.

8 Add the pea ⑦

The pea is printed **inside** the chamber, barely touching the floor. After printing, your teacher pokes it loose so it can rattle around. That's why the pea is added **after** grouping, so the holes don't cut it away!

- [ ] Drag a **Sphere** onto the whistle and make it `8 × 8 × 8`.
- [ ] Shift-click the whistle. **Align**: **right** dot and **middle** dot.
- [ ] Click the pea by itself and press `←` **15 times**. Now it's right above the chamber.
- [ ] Lift it to `1.9` (yes, one point nine). It sinks into the barrel and disappears. That's correct!
- [ ] Press `Ctrl` + `A`, then `Ctrl` + `G` to group the pea with your whistle.



This time: **right** and **middle**

WHY 1.9?

The chamber floor is 2 mm thick. Lifting the pea to 1.9 makes it sink just **0.1 mm** into the floor. That tiny spot holds it in place while printing, and is easy to snap later.

9 Final check ✅

- [ ] It looks like the Parts Map from the top: barrel, mouthpiece and hanger ring.
- [ ] There's a thin slot at the end of the mouthpiece and a rectangle window next to the barrel.
- [ ] It's about `39 × 26 × 20` mm and sitting flat on the workplane.

10 Save it and hand it in 💾

- [ ] Check the name at the top left says **Whistle – First Name Last Initial**. Not right? Click it and fix it.
- [ ] TinkerCad saves by itself while you work. To be sure, click the **TinkerCad logo** (top left) to go back to your designs. Your whistle should be there with **your name on it**.
- [ ] Tell your teacher your whistle is ready! Your teacher will download it from TinkerCad and 3D print it for you. 🖨️
- [ ] **Done early?** Make it yours! Add your initials on the side of the barrel with the **Text** shape (make it stick out just 1 mm), or try a different color. Don't change the holes, or it might stop whistling!

🤔 Think about it

1. Which part of the whistle makes the sound? Explain it in your own words.

2. If you made another whistle, what would you change?

CREDITS

This guide is adapted from "3D Printing Whistle" by **Ally Zhao** on Instructables, used under a Creative Commons BY-NC-SA license. Thank you, Ally! **Changes:** the original was made in 123D Design; this version rebuilds the same whistle (barrel, mouthpiece, air tunnel, sound window, hanger and pea) with TinkerCad shapes, using the original's sizes, and is rewritten as a step-by-step classroom worksheet with new drawings. It leaves out the original's tapered edge inside the sound window. This adapted version is shared under the same license, CC BY-NC-SA 4.0: free to share and change for non-commercial use, as long as you give credit and share alike. TinkerCad is a trademark of Autodesk; this guide is not made by or affiliated with Autodesk or Instructables.